AZURE INTEGRATION DOCUMENTATION

HR Management & Workforce Analytics Platform

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1. Azure Integration Objective

As per project requirements, cloud services must:
• Be used only where justified
• Support enterprise scalability
• Avoid heavy lab dependency
• Be conceptually integrated

Therefore, Azure integration was strategically designed for:
1. Identity Management (Azure Active Directory)
2. Serverless Automation (Azure Functions)
3. Optional Cloud Hosting Architecture

The objective was NOT to migrate the entire application to the cloud,
but to demonstrate enterprise readiness, scalability, and security enhancement.

2. Azure Active Directory (Secure Identity)

2.1 Business Justification

In an enterprise HR system, employee data is highly sensitive and subject to compliance requirements.
Leave approval workflows involve hierarchical authorization.
Therefore, robust authentication and role-based authorization mechanisms are mandatory.

Azure Active Directory (Azure AD) was conceptually integrated to provide:
• Enterprise-grade authentication
• Single Sign-On (SSO)
• Centralized identity management
• Role-based access control (RBAC)

• Secure JWT token-based API protection

2.2 Authentication Architecture Diagram

User (HR / Manager / Admin)
↓
Frontend Web Application
↓
Azure AD Authentication Endpoint
↓
JWT Token Issued (Access Token)
↓
Backend API Validates Token
↓
Role-Based Authorization Applied
↓
Secure Resource Access Granted

2.3 Role Mapping Table

Role	Access Level	Permissions
HR Executive	Operational	Create Employee, Manage Records
Manager	Approval	Approve/Reject Leave
Admin	Full Control	System Configuration & Full Access

2.4 Step-wise Conceptual Implementation

Step 1 – Register Application in Azure Portal
• Create App Registration
• Generate Application (Client) ID
• Configure Redirect URI
• Generate Client Secret

Step 2 – Define Role-Based Access Control (RBAC)

• HR Executive
• Manager
• Admin

Step 3 – Backend Configuration
• Install Microsoft.Identity.Web package
• Configure authentication middleware
• Enable JWT Bearer authentication
• Validate access tokens

Step 4 – Implement Authorization Policies
• HR → Create/Update Employee
• Manager → Approve Leave
• Admin → Full Administrative Access

2.5 Why Azure AD is Justified

Azure AD is justified because:
• Authentication is a core enterprise requirement
• It supports scalability to 10,000+ users
• It integrates with ERP systems
• It ensures compliance and audit readiness
• It centralizes identity management

3. Azure Functions (Serverless Logic)

3.1 Business Scenario – Leave Escalation Automation

In the HR system, leave approvals may remain pending due to managerial delays.
Manual tracking is inefficient and prone to oversight.

Business Rule:
If Leave Status = 'Pending' AND Approval Time > 72 hours,
then automatically escalate to senior authority.

3.2 Serverless Workflow Diagram

Leave Request Created

↓
Stored in SQL Server Database
↓
Azure Function (Timer Trigger - Every 6 Hours)
↓
Query Pending Leaves
↓
Check Time Condition (>72 hrs)
↓
Send Escalation Email Notification
↓
Update Leave Status to 'Escalated'

3.3 Implementation Steps

Step 1 – Create Azure Function App
• Choose Timer Trigger
• Schedule execution every 6 hours

Step 2 – Database Connectivity
• Use secure connection string
• Query leave records where Status = 'Pending'

Step 3 – Apply Business Logic
• Compare current date with leave request date
• If difference > 72 hours → escalate

Step 4 – Email Notification
• Integrate SendGrid or SMTP service
• Notify senior authority

Step 5 – Update Database
• Mark leave record as 'Escalated'

3.4 Cost Optimization Analysis

Serverless Feature	Enterprise Benefit
Pay-per-Execution	No cost when function is idle

Auto Scaling	Handles workload spikes automatically
Event-Driven Model	Efficient background processing
Managed Infrastructure	Reduced operational overhead

4. Optional Azure Hosting Architecture

Although not mandatory, the system can be deployed fully in Azure environment as follows:

Frontend (Browser)
↓
Azure App Service (Host ASP.NET Core API)
↓
Azure SQL Database (Managed Database Service)
↓
Azure Functions (Automation & Background Jobs)
↓
Azure AD (Authentication & Authorization)

4.1 Cloud Component Mapping Table

Azure Service	Purpose in HR System
Azure App Service	Host Backend API
Azure SQL Database	Managed Relational Database
Azure Functions	Background Automation
Azure AD	Identity & Access Management

5. Security & Compliance Considerations

• HTTPS encrypted communication
• JWT token validation in API middleware
• Role-based authorization enforcement
• Azure SQL firewall rules

• Logging & monitoring for audit trail

6. Enterprise Scalability Scenario

If organization expands from:
• 5 offices → 50 offices
• 200 employees → 20,000 employees

Azure ensures:
• Identity scaling via Azure AD
• High availability through managed services
• Disaster recovery capabilities
• Geo-distribution support for multi-region deployment

7. Conclusion

The Azure integration within the HR Management System is minimal, justified, and enterprise-aligned.
It strengthens authentication security, enables automated workflows,
supports scalability, and prepares the system for future ERP integration.